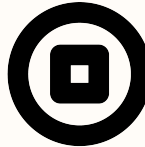
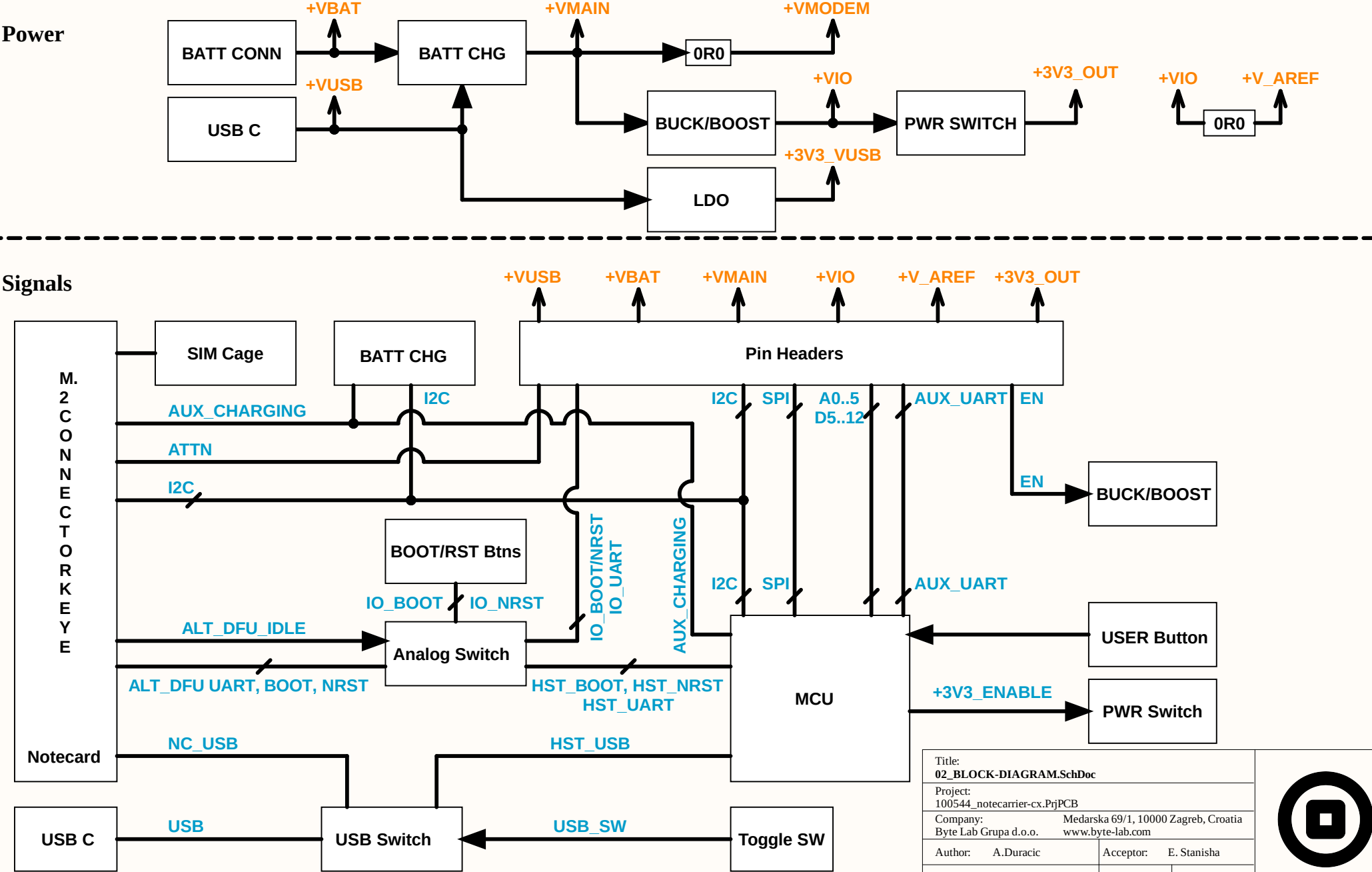
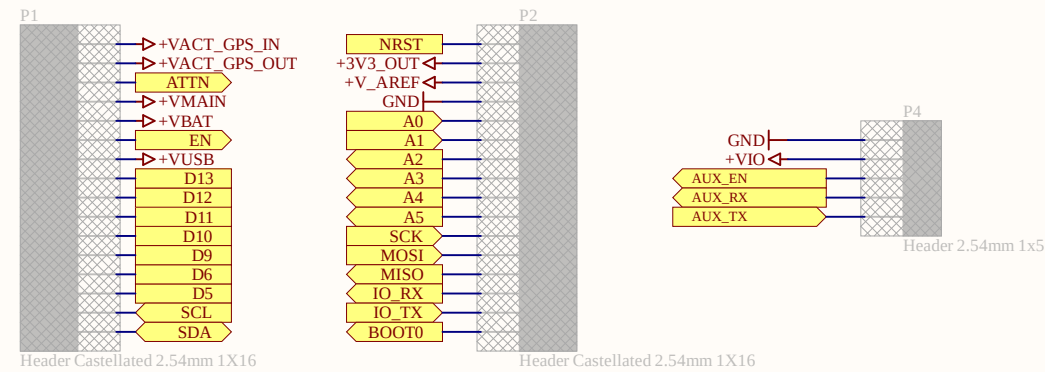


1		2		3		4				
Module name Notecarrier-CX Index 01 Cover 02 Block Diagram 03 IO Connectors 04 Notecard Connector 05 Power 06 MCU				REVISION BLOCK						
				Rev.		Date	Description	# VAR	# BOM	# ECO
				Proto.	Prod.					
				10	-	2025-4-23	Initial documentation release	-	3000-942-001	
				11	-	2025-6-9	Fixed USB, UART, added USER LED, swapped load SW Used shorter capacitors to increase clearance	-	3000-985-001	
				12	-	2025-7-30	Added AUX_UART pins, 0603 capacitors for clearance Flipped header pinout, added user button, fixed CHG_LED blinking	-	3001-001-001	
				12	-	2025-8-19	Added 1M pull-down to ENABLE_3V3, swapped user button logic	-	3001-002-001	
				13	-	2025-9-22	Switch S1 swapped, J1 connector vertical, HOST connection default	-	3001-018-001	
				14	-	2025-10-07	Switch S1 replaced, serigraphy adjusted	-	3001-026-001	
				15	-	2025-09-01	Added pull up/down to SWD, added PA15 to the AUX_CHARGING, added VMAIN sense, swaped MOSI/MISO	-	-	
				16	-	-	-			
				17	-	2026-02-12	Moved P4 in layout to align with 2.54mm grid			
				LIST OF AVAILABLE VARIANTS						
				# VAR		Variant name		Description		
REVISION FORMAT EXPLANATION										
-	-	202y-mm-dd	Development phase, date only		-					
10	-	202y-mm-dd	Prototyping phase, revision numeric		3000-xxx-xxx					
-	A	202y-mm-dd	Production phase, revision alphanumeric		3000-xxx-xxx					
Design considerations				Board support		Title: 01_COVER.SchDoc Project: 100544_notecarrier-cx.PrjPCB Company: Byte Lab Grupa d.o.o. Medarska 69/1, 10000 Zagreb, Croatia www.byte-lab.com Author: A.Duracic Acceptor: E. Stanisha Sheet 1 of 6 Rev: 17 Format: A4 				
SOFTWARE CONFIG NOTE: - Contains important notes for software development. - Things like: pin configuration, timing requirements, IC configuration etc.		DESIGN NOTE: - Hardware notes		PCB1 PCB 2201-080 DOC1 HW-E 100544						
1		2		3		4				

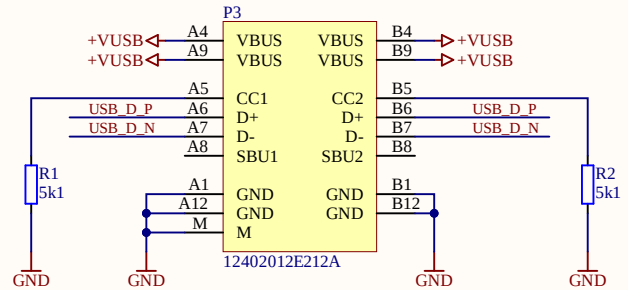


I/O Pin Headers



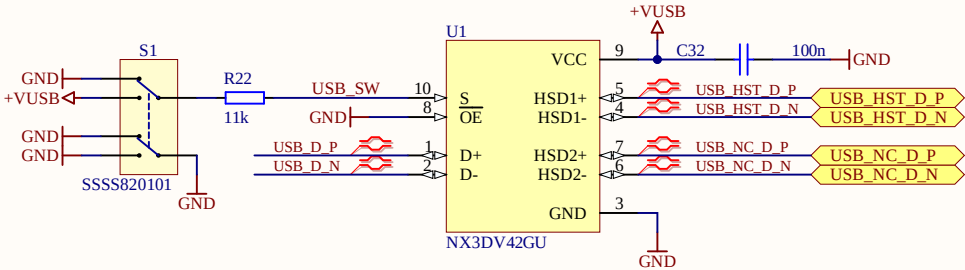
USB C Connector

DESIGN NOTE:
The USB C upstream facing port is configured to use 1.5A@5V or 3A@5V. The total power is limited by the downward facing port the device is connected to, which may be as low as 0.5A@5V.

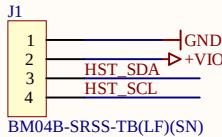


USB Switch

DESIGN NOTE:
The USB connection is routed depending on the position of SW1
NC (with marker): Host MCU
NO (w/o marker): Notecard

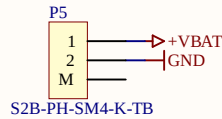


QWIIC Connector



DESIGN NOTE:
The I2C bus is pulled up by the Notecard

Battery Connector

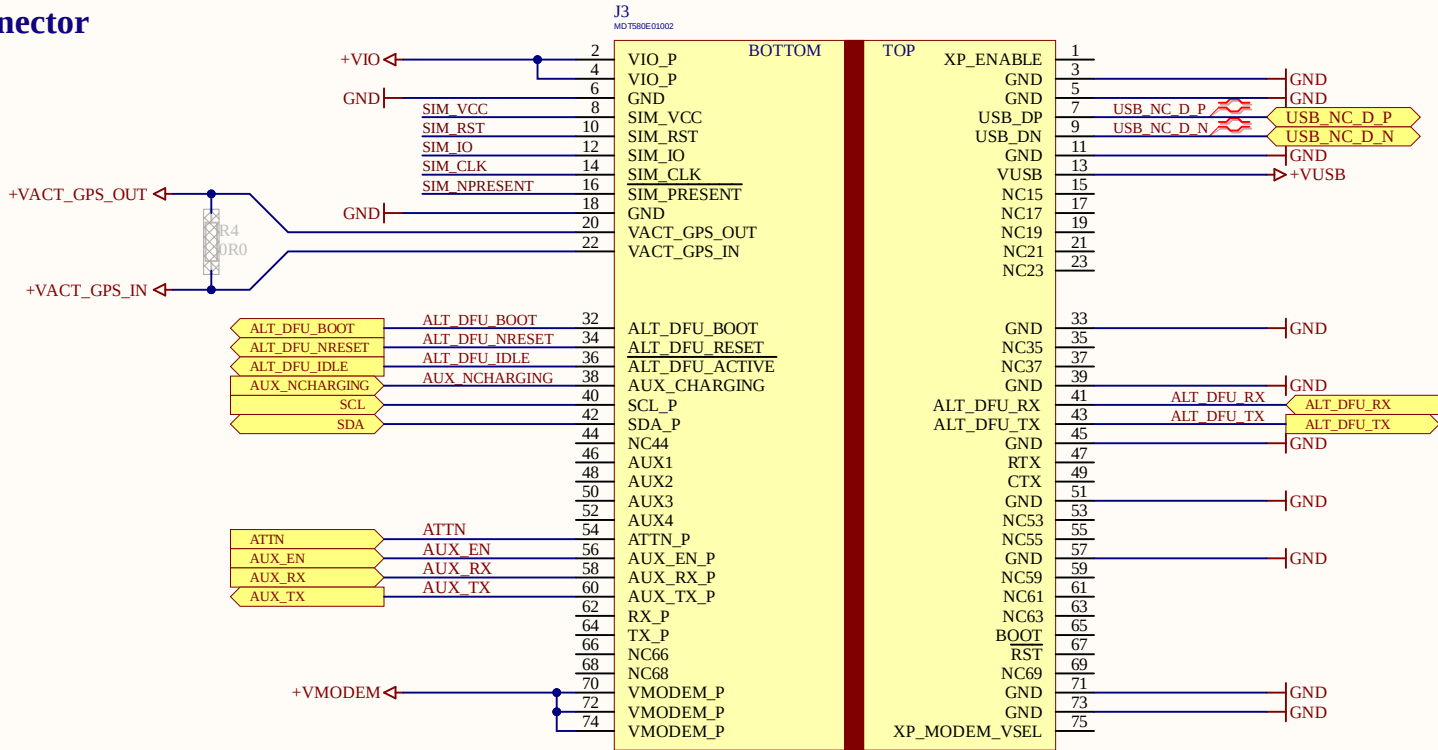


DESIGN NOTE:
Li-Po Battery
3.3V - 4.2V

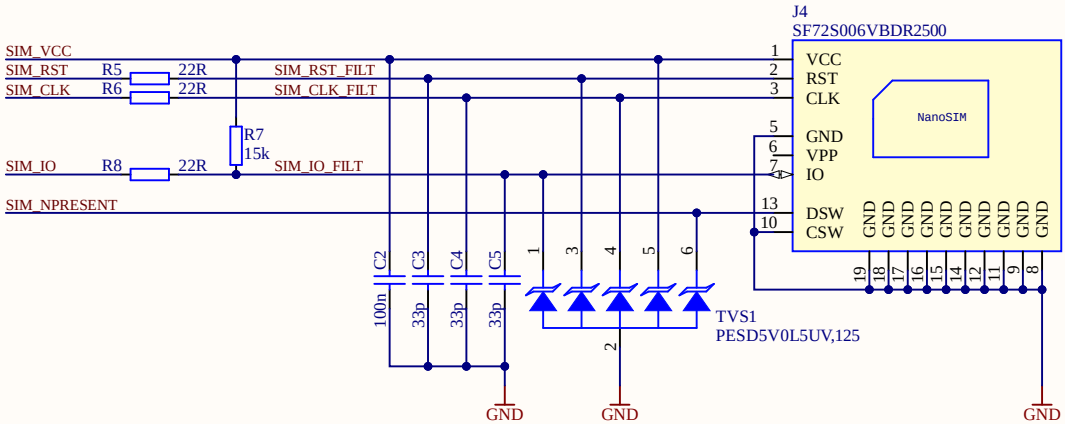
Title: 03_IO-CONNECTION.SchDoc		
Project: 100544_notecarrier-cx.PrjPCB		
Company: Byte Lab Grupa d.o.o.		Medarska 69/1, 10000 Zagreb, Croatia www.byte-lab.com
Author: A.Duracic	Acceptor: E. Stanisha	
Sheet 3 of 6	Rev: 17	Format: A4



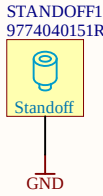
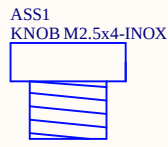
Notecard Connector



SIM Card



Notecard Support



Title: 04_NOTECARD-CONNECTION.SchDoc		
Project: 100544_notecarrier-cx.PrjPCB		
Company: Byte Lab Grupa d.o.o.		
Medarska 69/1, 10000 Zagreb, Croatia www.byte-lab.com		
Author: A.Duracic	Acceptor: E. Stanisha	
Sheet 4 of 6	Rev: 17	Format: A4



Battery Charger w/ Power Path

DESIGN NOTE:

Charger:
- Default charging voltage: 4.2V
- Default charging current: 320mA
- Default termination current: 20mA
- Default pre-charge current: 20mA
- No battery thermal protection
- No battery behaviour: +VBAT is pulled to VUSB - 0.7V through D4 & R20

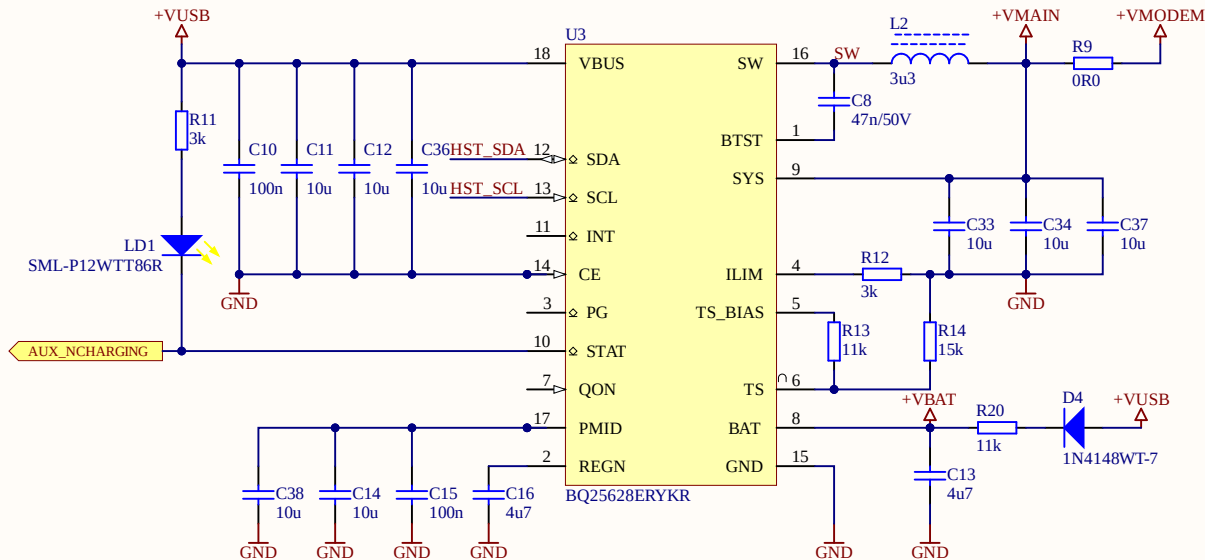
DESIGN NOTE:

Power Path:
- Input current limit: 3.5A
- 26mOhm reverse blocking transistor
- 15mOhm battery transistor

Buck Regulator:
- 1.5MHz frequency
- Input Voltage: 4.5-5.5V
- Output Voltage: 3.3-4.2V
- Peak current: 1.6A

SOFTWARE CONFIG NOTE:

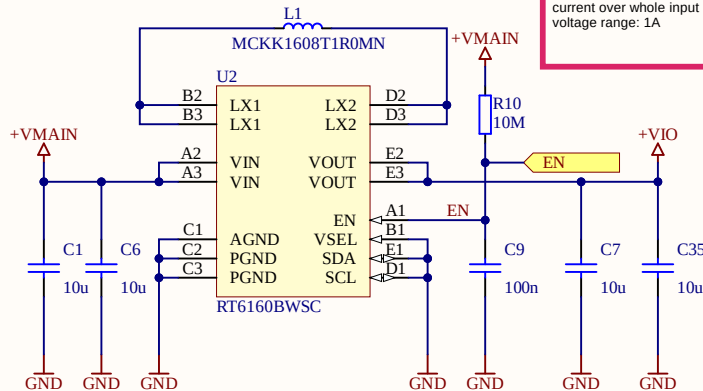
Charge current can be adjusted via I2C registers.
Make sure not to exceed 1.5A if programming the charge IC.



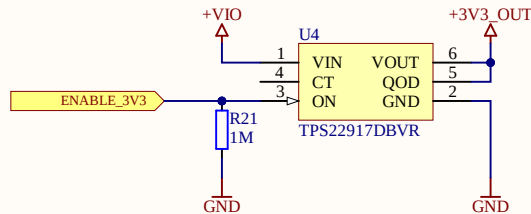
3V3 Buck/Boost Regulator

DESIGN NOTE:
Input Voltage: 2.5V - 5.5V
Output Voltage: 3.3V

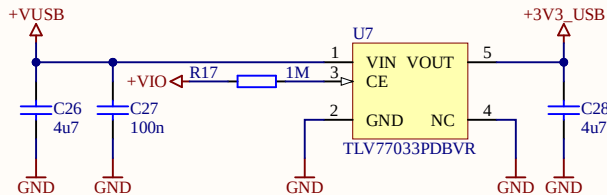
Max continuous output
current over whole input
voltage range: 1A



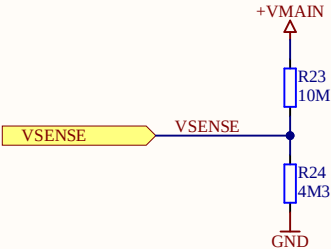
3V3_OUT Switch



USB LDO



VDETECT



Title:
05_POWER.SchDoc

Project:
100544_notecarrier-cx.PrjPCB

Company:
Byte Lab Grupa d.o.o. Medarska 69/1, 10000 Zagreb, Croatia
www.byte-lab.com

Author: A.Duracic

Acceptor: E. Stanisha

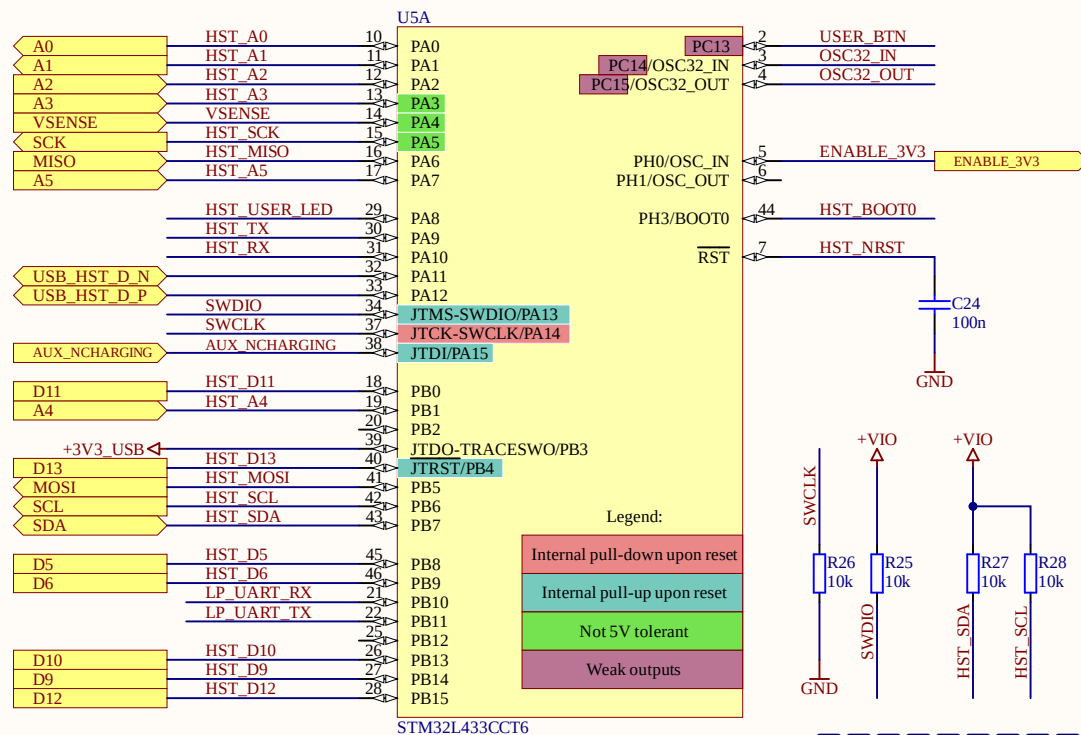
Sheet 5 of 6

Rev: 17

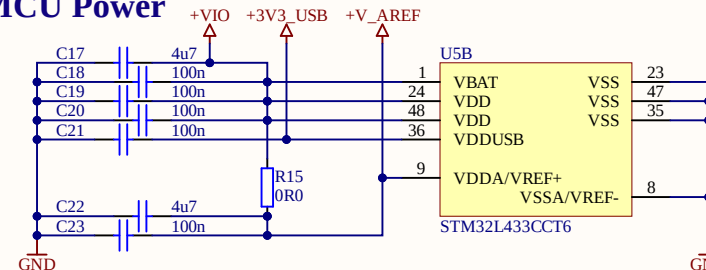
Format: A4



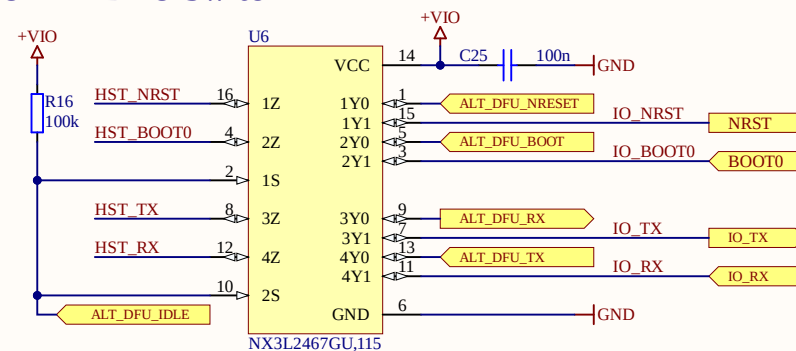
Host MCU



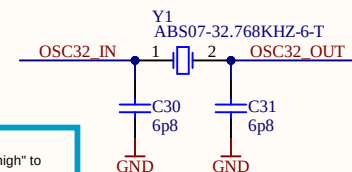
Host MCU Power



UART DFU Switch

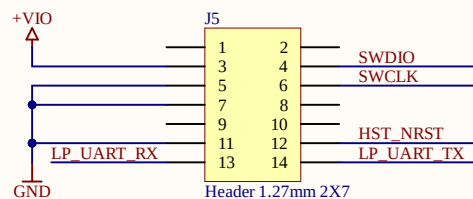


MCU Xtal

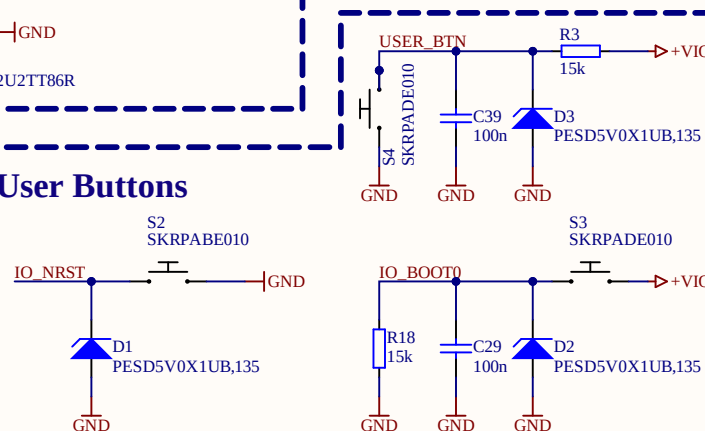


SOFTWARE CONFIG NOTE:
- Use drive level "medium high" or "high" to ensure startup

Debug Connector



User Buttons



Title: 06_MCU.SchDoc		
Project: 100544_notecarrier-cx.PrjPCB		
Company: Byte Lab Grupa d.o.o.	Medarska 69/1, 10000 Zagreb, Croatia www.byte-lab.com	
Author: A.Duracic	Acceptor:	E. Stanisha
Sheet 6 of 6	Rev: 17	Format: A4

